

ENCN445: Environmental Fluid Mechanics

Coastal Engineering Tutorials:

Module Three

Problem One:

A wave is travelling through a 2000 m deep ocean with $\omega = 0.495 \text{ rad/s}$, $k = 0.025 \text{ rad/m}$, and $\eta_0 = 0.3 \text{ m}$.

A Closer to the shore the ocean depth has decreased (gradually) to only 100 m. Calculate the angular frequency, wavenumber, and phase velocity at this new depth.

At this shallower location the wave is a shallow water wave so we actually have to calculate some things. We know that the angular frequency of the wave is conserved so

$$\omega_1 = \omega_2 = 0.495 \text{ rad/s}$$

where subscript 1 represents the open ocean value and subscript 2 represents the shallow value. The shallow wavenumber can be calculated with the dispersion relation. If this was a pure shallow water wave we would have

$$k_s = \frac{\omega}{\sqrt{gh_2}} = \frac{0.495}{\sqrt{9.81 \times 100}} = 0.0158 \text{ rad/m}$$

In reality this is a transitional wave so we need to solve

$$\frac{\omega^2}{g} = k \tanh(kh); \quad \frac{0.495^2}{9.81} = 0.0250 = k \tanh(kh)$$

Using our shallow water approximation for k this would give

$$0.0250 = 0.0158 \times \tanh(100 \times 0.0158) = 0.0145$$

This is clearly not a good enough estimation of k_2 so we will iterate knowing that the value is likely to be closer to the deep water limit k_1 as $\tanh(k_1 h_2) = 0.987 \approx 1$.

For $k_2 = 0.0253 \text{ rad/m}$ we get

$$k \tanh(kh) = 0.0253 \times \tanh(2.53) = 0.250 = \frac{\omega^2}{g}$$

This process shows that even for moderate values of kh the limit $\tanh kh \rightarrow 1$ as $kh \rightarrow \infty$ can be used.

The phase velocity is given by

$$c_2 = \frac{\omega_2}{k_2} = \frac{0.495}{0.253} = 19.5 \text{ m/s}$$

B *Finally, calculate the properties of this wave once it has reached a fluid depth of 4 m. This time calculate the same properties as in B but also calculate the new wavelength, the amplitude and the group velocity of this and the original wave.*

Subscript 3 refers to the new wave conditions.

Once again, the angular frequency is unchanged so

$$\omega_3 = 0.495 \text{ rad/s.}$$

At this depth we really do expect our wave to be a shallow water wave. If this is true then we can use the shallow water wave speed equation ($c = \sqrt{gh}$) to write:

$$k_3 = \frac{\omega}{\sqrt{gh_3}} = \frac{0.495}{\sqrt{9.81 \times 4}} = 0.0790 \text{ rad/m.}$$

We can check this with the full dispersion relation that we solved for in B:

$$\begin{aligned} \frac{\omega^2}{g} &\approx k_3 \tanh(k_3 h_3) \\ \frac{0.495^2}{9.81} &\approx 0.0790 \tanh(0.0790 \times 4) \\ 0.0250 &\approx 0.0242 \end{aligned}$$

This is close enough for us to use the shallow water equations without a significant loss of accuracy.

The phase velocity can be calculated as before:

$$c_3 = \frac{\omega_3}{k_3} = \frac{0.495}{0.0790} = 6.27 \text{ m/s}$$

Note that the phase velocity has decreased by more than a factor of two since location 2.

The wavelength can be calculated from the wavenumber in the usual way:

$$\lambda_3 = \frac{2\pi}{k_3} = \frac{2\pi}{0.0790} = 79.5 \text{ m}$$

We can see that our usual shallow water requirement of $h/\lambda < 0.04$ hasn't quite been met which is why there was some error in our check based on the dispersion relation.

Again, treating the wave as a shallow water wave, we can calculate the shallow water group velocity.

$$c_{g,3} = c_3 = 6.27 \text{ m/s}$$

To calculate the wave amplitude we also need the original group velocity. Using the deep water relationship we have

$$c_{g,1} = \frac{1}{2}c_1 = \frac{\omega_1}{2k_1} = \frac{0.495}{2 \times 0.025} = 9.9 \text{ m/s}$$

Now based on conservation of energy flux we have

$$\eta_3 = \eta_1 \sqrt{\frac{c_{g,1}}{c_{g,2}}} = 0.3 \sqrt{\frac{9.9}{6.27}} = 0.377 \text{ m}$$

Problem Two:

Waves are approaching a beach with a wavelength of 150 m and an amplitude of 0.7 m. The ocean depth is 4 m. The waves are approaching bathymetric contours (lines of constant depth) at an angle of 15 degrees (or $\pi/12$ radians.)

A Calculate the angle that the waves will be at relative to the bathymetric contours when the ocean is 3 m deep.

We will use subscript 4 and subscript 3 to refer to properties at 4 m and 3 m depth respectively.

First we need to calculate some wave properties at 4 m depth so that we can calculate the equivalent quantities at 3 m depth. The ratio $h/\lambda = 0.027 < 0.04$ so we will treat these as shallow water waves. The wavenumber can be calculated as

$$k_4 = \frac{2\pi}{\lambda_4} = \frac{2\pi}{150} = 0.0419 \text{ rad/m.}$$

The angular frequency can be calculated as

$$\omega_4 = k_4 \sqrt{gh_4} = 0.0419 \sqrt{9.81 \times 4} = 0.262 \text{ rad/s.}$$

The frequency of waves is conserved so

$$\omega_3 = \omega_4 = 0.262 \text{ rad/s.}$$

We calculate the wavenumber from the shallow water dispersion relation

$$k_3 = \frac{\omega_3}{gh_3} = \frac{0.262}{9.81 \times 3} = 0.0483 \text{ rad/m}$$

and the corresponding wavelength

$$\lambda_3 = \frac{2\pi}{k_3} = \frac{2\pi}{0.0483} = 130 \text{ m}$$

Finally, we can calculate the angle that the waves are approaching the shore at 3 m depth from trigonometry and the change in the wavelength:

$$\begin{aligned}\sin(\alpha_3) &= \frac{\lambda_3 \sin(\alpha_4)}{\lambda_4} = \frac{130 \sin(\pi/12)}{150} = 0.224 \\ \alpha_3 &= \sin^{-1}(0.224) = 12.6^\circ\end{aligned}$$

B Calculate the wave amplitude when the ocean is 3 m deep.

To calculate the wave amplitude we need to use conservation of energy flux. For this we need the group velocity at each location. Since both of these are shallow water waves we have

$$c_g = \sqrt{gh}$$

Using this we can calculate the change in wave amplitude between the two locations as

$$\begin{aligned}\eta_3 &= \eta_4 \left(\frac{c_{g,4}}{c_{g,3}} \right)^{1/2} \left(\frac{\cos(\alpha_4)}{\cos(\alpha_3)} \right)^{1/2} \\ &= \eta_4 \left(\frac{h_4}{h_3} \right)^{1/4} \left(\frac{\cos(\alpha_4)}{\cos(\alpha_3)} \right)^{1/2} \\ &= 0.7 \left(\frac{4}{3} \right)^{1/4} \left(\frac{\cos(15^\circ)}{\cos(12.6^\circ)} \right)^{1/2} \\ &= 0.7 \times 1.075 \times 0.9949 = 0.749 \text{ m}\end{aligned}$$

C What are the refraction and shoaling coefficients for the above situation?

The refraction coefficient is the second transformation term in B:

$$K_R = \left(\frac{\cos(\alpha_4)}{\cos(\alpha_3)} \right)^{1/2} = 0.9949$$

The shoaling coefficient is the first transformation term in B:

$$K_S = \left(\frac{c_{g,4}}{c_{g,3}} \right)^{1/2} = 1.075$$

D There is a small jetty at the beach. Local users want to protect the jetty with a breakwater placed where the water depth is 3 m. There is concern about the interaction of reflected waves with the original wavefield for the safety of small boats however. Two designs have been proposed: The first is a concrete seawall with an estimated reflection coefficient of 0.7 and the second is a rock breakwater with a reflection coefficient of 0.5. The concrete seawall would be preferred by the local group as it is cheaper. If the reflected waves need to have an amplitude less than 0.5 m for safety reasons which structure should be built. You

can assume that the wave conditions above are typical.

The reflected wave height in the case of the concrete seawall will be

$$\eta_r = \eta_i K_r = 0.749 \times 0.7 = 0.524 \text{ m}.$$

For the rock breakwater the reflected wave height will be

$$\eta_r = \eta_i K_r = 0.749 \times 0.5 = 0.375 \text{ m}.$$

Thus, the rock breakwater will meet the safety requirements but the concrete seawall will not.

E *Comment on how long the breakwater should be in relation to the area that needs to be protected. Are there any ways to reduce the length of the breakwater without impacting on its efficacy?*

Diffraction will cause the wave field to bend around the edge of the breakwater. To account for this the breakwater needs to be longer than the region being protected.

If the designers wanted to reduce the length of the breakwater they would need to bring the structure closer to the shore so that there was less distance for the waves to diffract before reaching the shore. This could be done by moving the entire structure or by adding an extra section perpendicular to the main wall and towards the shoreline. Waves will diffract from the end of the breakwater which is now closer to the shore.

It is not immediately clear whether it is better to build an extension towards the shore or to move the entire breakwater closer to the shoreline. Extending the breakwater will involve the construction of an additional length of wall so will have an associated cost. Bringing the entire breakwater closer to the shore will change the typical wave amplitude that the breakwater observes. This could mean that the reflected waves are now too large for the safety requirements in D. It could also mean that the breakwater will have to sit higher above the water line. This height will be offset by a reduced water depth however so the total height of the wall (from the bottom of the sea) might actually be less.

Problem Three:

A keen surfer and fluid dynamicist is visiting a new area and wanting to know which local beach is most likely to have the best waves. From a fluid dynamics point of view, this corresponds to plunging waves rather than spilling or surging waves. Whereas most people would just use the internet or ask locals, this particular surfer wants to use their knowledge of fluid dynamics to determine where to go surfing, thus satisfying both of their passions. There are three local beaches with bed slopes of 0.8° , 1.6° , and 4.5° respectively. The waves that are approaching this section of beach have a deep water wavelength of 200 m and a deep water wave height of 1 m.

A *Determine which beach is most likely to have plunging waves based on the above characteristics*

Plunging waves are found when

$$0.4 < \zeta = \frac{\tan \beta}{\sqrt{H_d/\lambda_d}} < 2.0.$$

For the first beach we have

$$\zeta = \frac{\tan \beta}{\sqrt{H_d/\lambda_d}} = \frac{\tan 0.8}{\sqrt{1/200}} = 0.20,$$

for the second beach we have

$$\zeta = \frac{\tan \beta}{\sqrt{H_d/\lambda_d}} = \frac{\tan 1.6}{\sqrt{1/200}} = 0.40,$$

and for the third beach we have

$$\zeta = \frac{\tan \beta}{\sqrt{H_d/\lambda_d}} = \frac{\tan 4.5}{\sqrt{1/200}} = 1.1.$$

Thus the third beach is most likely to have plunging breakers. The first beach will have spilling breakers and the second is at the transition between plunging and spilling breakers.

B Calculate how far out the surfer will have to paddle to get to the location where the waves are breaking. Assume that the waves are approaching the beach directly (no refraction), the beach slope is constant, our wave solution is valid where the waves are breaking, and a value for the breaking index κ of 1.15. You can also assume that where the wave is breaking it is a shallow water wave.

We need to find the location where $\kappa = H/h = 1.15$. This is first going to require calculating how shoaling will affect the wave. In the deep ocean we know that $\lambda = 200\text{ m}$. Therefore our deep water wave properties are:

$$\begin{aligned} k_1 &= \frac{2\pi}{\lambda_1} = 0.0314\text{ rad/m} \\ \omega &= \sqrt{gk} = \sqrt{9.81 \times 0.0314} = 0.555\text{ rad/s} \\ c_{g,1} &= \frac{1}{2} \sqrt{\frac{g}{k}} = \frac{1}{2} \sqrt{\frac{9.81}{0.0314}} = 8.84\text{ m/s}. \end{aligned}$$

where subscript 1 refers to the deep water wave properties and subscript 2 will refer to the shallow water wave properties.

We know that

$$\begin{aligned} \kappa &= 1.15 = \frac{H_2}{h_2}, \\ H_2 &= \frac{H_1 c_{g,1}^{1/2}}{c_{g,2}^{1/2}}, \\ c_{g,2} &= g^{1/2} h_2^{1/2}. \end{aligned}$$

Combining all of these together yields:

$$\begin{aligned}\kappa &= 1.15 = \frac{H_1 c_{g,1}^{1/2}}{g^{1/4} h_2^{5/4}}, \\ h_2 &= \left(\frac{H_1 c_{g,1}^{1/2}}{1.15 g^{1/4}} \right)^{4/5} = \left(\frac{1 \times 8.84^{1/2}}{1.15 \times 9.81^{1/4}} \right)^{4/5} = 1.35.\end{aligned}$$

Thus, the waves will break at a ocean depth of 1.35 m. Calculating how far the surfer will need to paddle is now a matter of simple trigonometry:

$$d = \frac{h_2}{\tan \beta} = \frac{1.35}{\tan 4.5} = 17.2 \text{ m}$$